

Lab 13 Worksheet — RAM Identification, Installation and Channel Configuration

Name: ____ Date: ____

Exam objective: Identify RAM types and form factors, install modules correctly and configure dual-channel operation, verifying the result in firmware and the OS (Core 1 objective 3.3).

Record as you go

Step	What you did	What you observed
1	Put on the anti-static strap, power down the machine, disconnect it from the mains and hold the power button for 15 seconds to discharge.	
2	Remove a memory module by pressing the retaining clips outward at both ends simultaneously, then lift the module by its edges without touching the gold contacts.	
3	Read the module label and decode every field: capacity, DDR generation, speed rating, CAS latency and whether it is ECC or non-ECC.	
4	Identify the form factor: a full-length DIMM for desktops and servers, or a shorter SODIMM for laptops and small-form-factor systems.	
5	Locate the notch in the module's contact edge and confirm it aligns with the key in the slot — DDR3, DDR4 and DDR5 are keyed differently and cannot be interchanged.	
6	Consult the motherboard manual or its silkscreen to identify which slot pairs form the dual-channel banks, typically slots 1 and 3 or the colour-matched pair.	
7	Install two matched modules into the correct channel pair, pressing straight down at both ends until the retaining clips snap closed by	

Step	What you did	What you observed
	themselves.	
8	Boot into BIOS/UEFI and record the total memory detected, the operating speed and whether the system reports single or dual channel.	
9	Boot into Windows, open Task Manager, go to Performance then Memory, and record total capacity, speed, form factor, slots used and hardware reserved.	
10	Compare the reported speed against the module's rated speed and explain any difference — a module rated above the chipset's supported speed runs at the lower rate unless XMP is enabled.	
11	Explain what ECC memory does, that it makes a system more stable rather than faster, and why it requires explicit CPU and motherboard support.	
12	Write a memory upgrade recommendation for a stated workload, naming the capacity, generation, speed and slot configuration, and justify it against the chipset's maximum.	

Verification

Success criterion: The system POSTs and reports the full installed capacity, Task Manager confirms the correct speed and slot usage, dual channel is confirmed in firmware, and your upgrade recommendation names a specific module that the chipset actually supports.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
